output. Finally, for word-specific constraints, we compute the weight using Eq. (2) and additionally perform a small grid search over neighboring values on a held-out set of examples to fine-tune the steering effect. The steering vector $c\mathbf{u}_l$ is then added to the corresponding residual stream layer and the forward pass is resumed with the updated residual stream value $\tilde{\mathbf{x}}'_l = \mathbf{x}'_l + c\mathbf{u}_l$. This procedure is carried out at a single layer across all token positions, motivated by previous findings that show models tend to deviate from instructions as they generate more tokens (Li et al., 2024). To find the optimal layer for steering, we perform a sweep across a subset of the model's layers, measuring the effect on a held-out set of queries and performing a perplexity-based quality check. Details about this procedure are provided in Appendix E.

2.3 Experimental Setup

Data. We use an augmented version of the IFEval dataset (Zhou et al., 2023a), which consists of 25 distinct instructions, each paired with multiple base queries and expressed in different phrasings, for a total of 541 prompts. To evaluate format instructions, we focus on a subset of 163 examples that specifically relate to the output format. This subset includes instructions such as "The entire output should be wrapped in JSON format," requests to use a particular language (e.g., "Please respond using only the Punjabi language"), formatting requirements like "Wrap your entire response with double quotation marks," and casing instructions (e.g., "Answer using lowercase letters only"). A complete list of the instructions used is provided in Appendix B.[3] For length instructions, we generate prompts by concatenating base queries from IFEval to instructions derived from templates such as "Answer using at most $\{n\}$ sentences." For word-specific instructions, we use a subset of IFEval containing instructions about the inclusion or exclusion of keywords. The subset has 203 examples, and each example contains a prompt with a single keyword-related instruction. For steering vector computation and layer selection, we construct a separate set of synthetically generated prompts by combining base queries from IFEval with corresponding instruction descriptions to avoid test information leakage. Additional details about the data used are provided in Apps. C and D.

Metrics. To quantify the models' adherence to format instructions, we compute the "loose" instruction-following accuracy using the official IFEval evaluation script. For length instructions, we count the number of words or sentences in the model's output. For word-specific constraints, we verify the presence or absence of the specified keywords in the model's response, again using the IFEval evaluation script. We assess the statistical significance of differences in average instruction-following accuracy with and without steering using McNemar's test (McNemar, 1947). Results where steering leads to a significant improvement (p-value < 0.01) are marked with an asterisk (*).

In addition to assessing the model's instruction-following capabilities, it is important to verify that the model still effectively addresses the base query, even when generating under constraints. To measure the overall quality of the response, we set up a GPT-4o-based evaluation. For each base query x, GPT-4o generates a set of five yes/no questions aimed at assessing the quality of the response to x. For instance, given the query "Write an essay about the history of Martin Van Buren's presidency," GPT-4o generates questions such as "Does the essay provide context on the political, social, and economic climate during Van Buren's presidency?" and "Is the essay written in clear, grammatically correct English, and does it follow a logical structure?". Next, given the original query x, a model's response to x, and the generated questions we prompt GPT-4o to answer each question, provide a rationale for each yes/no response, or reply with "N/A" if the question is not applicable. We compute the proportion of questions answered positively for a given query and average this score across all queries to obtain a *response quality score*. We repeat this experiment 3 times and report the mean and standard error. Additional details are provided in Appendix F.[4]

Evaluation. We conduct experiments using the instruction-tuned versions of Phi-3 Mini (Abdin et al., 2024), Gemma 2 2B, 9B (Gemma Team, 2024), and Mistral 7B v0.1(Jiang et al., 2023). In addition, we investigate the transferability of steering vectors from the instruction-tuned to the base versions of Gemma 2 2B and 9B. All models are evaluated in a zero-shot setting, with outputs

[3] Although they are not strictly related to format, we include language constraints (e.g., "the answer should be in German") in this group, as, like other formatting instructions, they modify the surface-level presentation of the output without affecting the underlying content.

[4] Note that while the model's response may be influenced by specific instructions, these instructions are not provided to GPT-4o during the evaluation. The goal here is to focus the quality assessment solely on the comprehensiveness and relevance of the response with respect to the base query.

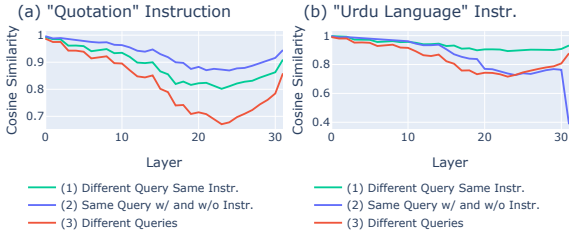

Figure 2: **Residual stream similarity across layers.** Phi-3's residual stream activations show higher cosine similarity between examples with the same instruction compared to those without, indicating effective capture of instruction-relevant features.

Table 1: **Steering Vector Projections.** The projection onto the vocabulary space of the contrastively-computed vectors promotes tokens that are semantically related to the respective instruction.

Instruction	Layer	Top Tokens
JSON Format	18	␣[{, ␣json, ␣`{, ␣JSON
Capitalize	28	␣PRO, ␣TH, ␣ FOR, ␣AND
Highlight Text	26	*`, ␣*, ␣*, (*, *\\
Lowercase	18	␣lower, ␣lowest, ␣russ
Bullet List	28	*), ␣•, *, ␣*), •, */
Quotation	26	␣`", ␣'", ␣"", ␣"; ␣"</
Urdu Language	24	␣Islam, ␣Pakistan, ␣Pak
Hindi Language	18	␣Indian, raj, ␣India
German Lang.	16	␣die, ␣im, ␣dies, ␣gener

decoded greedily. We assess our method's performance by steering the model under two input settings: (1) with text instructions provided in the input, and (2) without any text instructions. These settings allow us to explore two questions: Are the computed steering vectors informative enough to guide the model's behavior even without explicit instructions? And can steering further improve performance when instructions are provided in the input?

3 Format Instructions

Representational Analysis. To assess whether the model effectively captures instruction-related information in its internal representations, we analyze Phi-3's residual stream activations at the last token of the input by calculating cosine similarity across three sets of inputs: (1) pairs of inputs that share the same base query, one with and one without the instruction; (2) inputs with different base queries but the same instruction; and (3) inputs with different base queries and no instruction. High similarity in set (2) would suggest that the model captures a shared feature (the instruction), while we expect set (1) to show relatively high similarity due to shared base queries, and set (3) to show lower similarity due to the absence of any shared features. Figure 2 shows how these measures evolve across layers for two instructions. For the "quotation" instruction, where the model is asked to wrap the output in quotation marks, there is a clear difference between sets (2) and (3), indicating that the instruction is partially captured (green vs. red lines in Figure 2a). For the "Urdu language" instruction, set (2) shows higher similarity than set (1), suggesting strong representation of the instruction (green vs. blue lines in Figure 2b). These results indicate that the model can effectively encode instruction-relevant features at the last input token, which supports steering model behavior based on specific constraints. Additional representational analyses are provided in Appendix H.

Steering Vector Computation. We compute steering vectors for each of the 12 format-related instructions in the IFEval subset and for the 19 language-based instructions specified in the dataset. During the selection of the optimal steering later, we compare the validation score with and without steering to ensure that the steering intervention leads to an improvement. If no layer shows an improvement in the validation score, no steering is applied at test time. Details about the layers selected for steering are provided in Appendix E. To ensure that the steering vectors effectively capture the information related to the instructions, we inspect them by projecting the vectors onto the model's vocabulary space (nostalgebraist, 2020; Geva et al., 2022). We project the vectors using the model's unembedding matrix and examine the vocabulary tokens with the highest logit values. Table 1 presents the top tokens associated with several of the instructions we consider. We observe that the tokens promoted by the steering vectors are semantically related to the intended instruction, providing an initial validation that the vectors are capturing the desired features.

Steering Results. We first evaluate steering on inputs without explicit text instructions (Figure 3a). In this setting, the instruction-following accuracy without steering hovers around 10% as the input has no information about the instruction. This non-zero accuracy reflects cases where the models incidentally satisfy the instruction, such as when a model rephrases a sentence without using commas, thus accidentally meeting the "no comma" constraint. When steering is applied, we observe a consistent increase in instruction adherence across all models, with accuracy improving to approximately 30%. This shows that the steering vectors encode meaningful information of the instructions

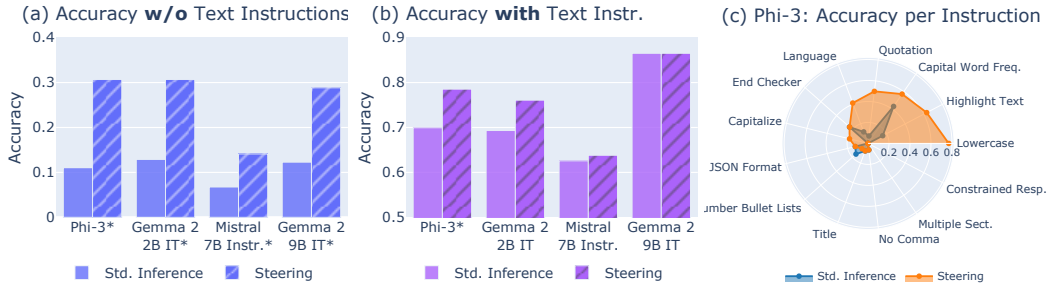

Figure 3: **Format Instructions.** (a) Instruction-following accuracy without explicit text instructions shows significant improvement with steering across all models. (b) Steering enhances accuracy even when text instructions are provided. (c) Per-instruction accuracy for Phi-3 without text instructions.

and can be effectively used to steer the models toward the intended behavior. Next, we evaluate on inputs with instructions provided in the text (Figure 3b). As expected, the instruction-following accuracy without steering is higher in this setting, ranging between 60% and 90%. Nevertheless, steering still results in a noticeable performance boost for two out of four models, demonstrating that steering can enhance instruction adherence even when the instructions are explicitly given.

To further analyze which instructions benefit most from steering, we break down Phi-3's performance by instruction on input without explicit text instructions (Figure 3c). Notably, we observe significant improvements for instructions such as "Lowercase," which requires the output to be entirely in lowercase characters, and "Highlight Text," which asks the model to emphasize parts of the response using markdown syntax. However, certain instructions display variability in steering effectiveness. For instance, the "End Checker" instruction requires the model to finish the response with a variable, input-dependent sentence (e.g., "Hope you agree with me," or "Is there anything else I can help you with?"). Because the steering vector is computed by averaging over all examples containing this instruction, it likely fails to capture the specific sentence required in each individual case, reducing its ability to steer the model toward precisely following this type of instruction.

Response Quality Evaluation. Figure 4 compares the average differences in response quality scores for the four models. The two leftmost groups of bars show the score changes due to steering, with and without input instructions (calculated only for cases where steering is applied). Satisfying specific constraints during generation may lead to less comprehensive responses and hence following instructions is expected to impact the response quality. To show this effect, we also present the differences in quality scores observed when explicit text instructions are added to the base queries without steering. We observe small decreases in quality score due to steering in the setting including instructions (with the exception of Gemma 2B) while the decreases in the setting without instructions are slightly larger but comparable to the effect caused by simply adding the instructions as text. However, we also observe instances where steering compromised the quality of generation–examples where the model generated nonsensical tokens or repeated itself. These failures likely result from suboptimal steering or partially insufficient objectives during the search for layers to intervene, as steering for specific properties can increase perplexity or reduce fluency (Turner et al., 2023; Stickland et al., 2024). We provide some examples from the few cases we observed in Appendix F.

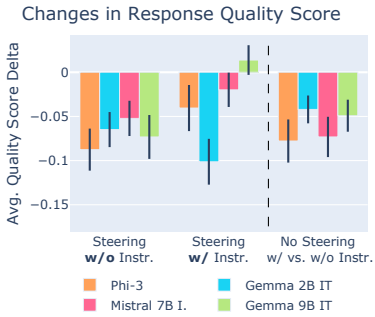

Figure 4: **Quality Score.** Average response quality score changes for four models, comparing the effects of steering with and without input instructions to the effect of adding instructions without steering. Steering decreases quality similarly to adding instructions.

4 Length Instructions

Steering Vector Computation. Length constraints can be specified in various ways, such as by the number of sentences, words, or lines. However, unlike format instructions, it seems impractical to compute a separate steering vector for each possible length constraint. Instead, we focus on captur-